

Practice Exam 3

Multiple Choice

1. What evidence strongly indicates (if any) that the suprachiasmatic nucleus (SCN) produces the circadian rhythm itself?

- A. The SCN does not produce the circadian rhythm
- B. SCN cells isolated from the body continue to produce a circadian rhythm
- C. Animals with a faster circadian rhythm have larger SCN
- D. The SCN increases its activity during wakeful periods and decreases it during sleep

2. Slow, large-amplitude EEG waves characterize ____ of sleep and is promoted by the ____?

- A. stages 1 & 2, hypothalamic system
- B. stages 1 & 2, basal forebrain
- C. stages 3 & 4, hypothalamic system
- D. stages 3 & 4, basal forebrain

3. People who are blind because of cortical damage can still synchronize their circadian rhythm to the local pattern of day and night.

- A. True
- B. False

4. Processes that reduce any discrepancies from the set point are known as ____.

- A. negative feedback
- B. positive feedback
- C. feedback loops
- D. feedback modifiers

5. How does hypovolemic thirst differ from osmotic thirst?

- A. Hypovolemic thirst is stronger
- B. Osmotic thirst is stronger
- C. Someone with hypovolemic thirst prefers slightly salty water
- D. Someone with hypovolemic thirst prefers pure water
- E. None of the above

6. Increased blood glucose causes increase of ____, which ____ the ability of glucose to enter the cells.

- A. insulin ... increases
- B. insulin ... decreases
- C. glucagon ... increases
- D. glucagon ... decreases

7. High levels of leptin are associated with ____.

- A. decreased activity and eating
- B. increased activity and eating
- C. decreased activity and increased eating
- D. increased activity and decreased eating

8. When Kasha has food placed in their mouth, they begin to salivate. The salivation behavior is called the ____.

- A. unconditioned stimulus
- B. conditioned stimulus
- C. unconditioned response
- D. conditioned response

9. Spatial memory is most strongly associated with what brain region?

- A. Hippocampus
- B. Parietal cortex
- C. Basal ganglia
- D. Prefrontal cortex
- E. Cerebellum

10. What is anterograde amnesia?

- A. Loss of factual memory
- B. Loss of memory for personal experience
- C. Loss of memory for space and time
- D. Inability to form new memories

11. What is true about Hebbian synapse?

- A. It can increase its effectiveness as a result of simultaneous activity in the presynaptic and postsynaptic neuron
- B. It can be either excitatory or inhibitory, depending on the activity of other nearby synapses
- C. It includes either an AMPA site or an NMDA site
- D. It can send messages between cells in either direction

12. According to the James-Lange theory, we experience emotion ____.

- A. before our actions
- B. simultaneously while acting
- C. after feeling autonomic arousal
- D. and need to process it before acting
- E. all of the above

13. Why do we know more about the brain mechanisms of fear and anxiety than we do about other emotions?

- A. Clinical psychologists have greater interest in anxiety than other emotions
- B. Anxiety depends on brain areas that are easier to reach surgically
- C. Unlike other emotions, anxiety depends on only a single neurotransmitter
- D. Researchers can more satisfactorily measure anxiety than other emotions in laboratory animals

14. Development of a male fetus depends on suppressing the development of the embryonic ____ and stimulating the development of the ____.

- A. Wolffian duct ... Müllerian duct
- B. Müllerian duct ... Wolffian duct

15. What is NOT a true statement?

- A. In males, sexual arousal is negatively correlated with testosterone levels
- B. In males, castration decreases sexual activity and sexual interest
- C. In females, fertility is highest during the middle of the menstrual cycle
- D. In males, testosterone primes areas to increase dopamine release

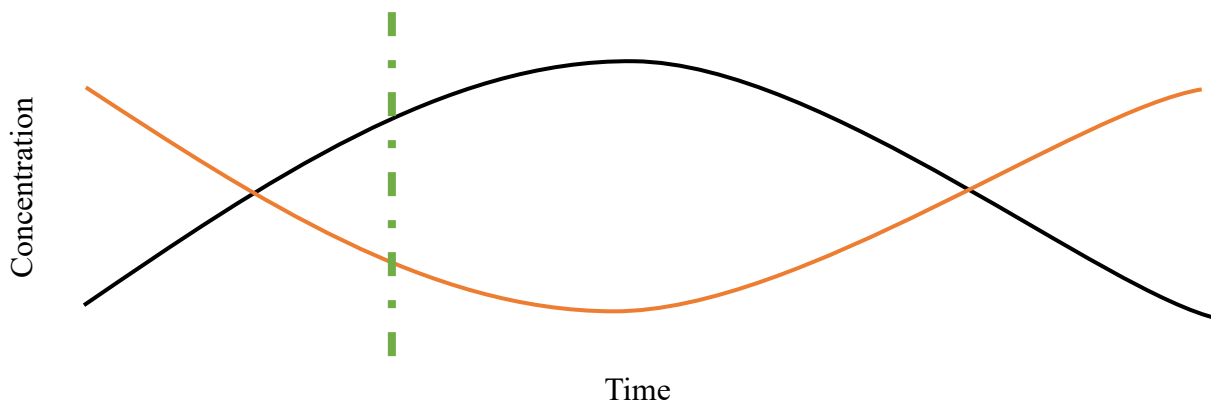
Open response

For each part, please answer using single words/phrases/ paragraphs/sketch as applicable.

Sleep

1. (8 points) How does the SCN influence our circadian rhythm? (Hint: retina, SCN, their proteins and mRNAs, melatonin)

2. (4 points) The SCN keeps track of light cycles using several different proteins and their mRNA. The following illustration represents the concentration of TIMELESS mRNA and protein over a period of time. If the **orange line** represents the concentration of **TIM mRNA**, **what** kind of behavior would you expect from an average human at the **green dotted line**? (Awake or sleeping)



What would happen to the TIM mRNA if this individual was awake and exposed to a bright, blue light?

3. (3 points) **Name** one sleep disorder. What is a known **cause** for this sleep disorder and what is a known **treatment** option for this sleep disorder?

Internal Regulation

1. (6 points) **How** do humans regulate their internal environment and maintain equilibrium? Go into detail using an example: response to heat or cold. (Hint: Include information on sensor, control center, effector).

2. (4 points) Marisa is sick and has a fever. **Explain** what a fever is and how this is good for Marisa.

3. (2 points) **Describe briefly** how bulimia nervosa resembles drug addiction.

Learning & Memory

1. (7 points) In each scenario below, identify the behavior that is being rewarded/punished and the

reward/punishment. Also, identify if it is an example of positive reinforcement, negative reinforcement, negative punishment, or positive punishment.

A) Xavier drives very carefully because he has gotten several traffic tickets in the past. (3pts)

B) Lynus has learned that if he leaves early for work, she can avoid heavy traffic. (3pts)

C) **What** type of learning is this in both scenarios? (1pt)

2. (2 points) **What** is the difference between procedural memory and episodic memory?

3. (8 points) The cellular mechanism of learning and memory is long-term potentiation. Please answer following:

A) **How** is long-term potentiation achieved? (Hint: Name and describe the three properties discussed for LTP) (6pts)

B) **What** two type of glutamate receptors are necessary for long-term potentiation? (2pts)

Emotional Behaviors

1. (4 points) **What** is the difference between the commonsense approach to emotion and the James-Lange Theory of emotion? (I'm happy to see drawings)

2. (6 points) Various experiments have helped us to understand how humans perceive emotion.

A) **What** brain region is thought to be essential for eliciting the arousal associated with, and for understanding fear? (2pts)

B) Please draw a brief diagram showing what brain regions are working together to create fear-based startle reflexes. (4pts)

Reproductive Behaviors

1. (6 points) **What** does the SRY gene do?

2. (3 points) **What** is a sensitive period? Give an example?

3. (3 points) **What** is the main difference between organizing effects and activating effects of sex hormones?